Zack Valavanis

Software Engineer | Python, FastAPI, React, TypeScript, PostgreSQL

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SUMMARY

Full-stack software engineer with five years of finance experience, building and deploying production web applications with Python, FastAPI, React, TypeScript, and PostgreSQL. Shipped 3 full-stack applications to production, including 1 generating revenue. Combines hands-on engineering with deep financial domain knowledge (FINRA Series 3, 7, 63, 65) in portfolio analytics, risk, and derivatives. Pursuing a B.S. in Computer Science at CU Boulder with a 4.0 GPA.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Ruby, VBA
Backend: FastAPI, SQLAlchemy, Pydantic, Node.js, Express, Ruby on Rails, REST APIs, JWT Auth, OpenAI API
Frontend: React, React Native, Vite, HTML, CSS
Databases & Cloud: PostgreSQL, Supabase (Row-Level Security), Vercel, Railway, Heroku
Testing & Tools: pytest, Jest, Git, GitHub, CI/CD, Tableau, Excel

PROFESSIONAL EXPERIENCE

Data Analytics Analyst / Client Associate | Neuberger Berman, Chicago, IL 2023 – Present
- Developed Python and VBA automation and data validation scripts for financial reporting, saving 56 hours of manual work per year.
- Managed a $2.7B diversified portfolio, building client dashboards on risk-adjusted returns, allocation, volatility, and benchmarks.

Real Estate Analyst | Accenture, Chicago, IL 2022 – 2023
- Managed a $1.1B portfolio of 37 commercial office properties, improving tenant satisfaction by 20%.
- Redesigned vendor operations and processes, generating $200K in savings and a 5% reduction in operating expenses.

Risk Analyst | Advantage Futures, Chicago, IL 2020 – 2022
- Automated quarterly data compilation with VBA macros, cutting reporting time by over 50%.
- Analyzed margin usage and open interest across futures, including Bitcoin and Ethereum, reducing risk exposure by 20%.

OTHER EXPERIENCE

Software Engineering Intern, Frontend | Capture (privacy-first social media app) Summer 2025
- Built and refined React Native screens and reusable UI components as one of three engineers on an early-stage team.

Founder & Frontend Engineer | Salus Wellness (AI personal training app) Summer 2025
- Built React Native screens from Figma prototypes, owning frontend implementation, state management, and debugging.
- Collaborated with a 7-person founding team, using AI-assisted development to accelerate delivery toward launch.

PROJECTS

AlgoLedger | Algorand Portfolio Analytics & Tax Platform Live Product
Python, FastAPI, React, TypeScript, PostgreSQL, Supabase, Railway, Vercel
- Launched a freemium Algorand analytics and tax tool with paying users, supporting 3 cost-basis methods (FIFO, LIFO, HIFO).
- Engineered a FastAPI backend on PostgreSQL with Row-Level Security for multi-tenant data isolation and API rate limiting.
- Grew to 200+ users through organic launches on Reddit and X, reaching the top post on r/algorand.

StreakCat | Habit Tracking Web Application Live Demo
Python, FastAPI, OpenAI API, SQLAlchemy, Pydantic, PostgreSQL, React, Vite, JWT, pytest, Railway, Vercel
- Built a full-stack habit tracker with a GitHub-style streak heatmap and a cat mascot that reacts to user consistency.
- Integrated an OpenAI chat assistant that manages tasks via natural language and auto-builds the user's calendar.
- Designed a FastAPI backend with 3 routers (auth, users, tasks), SQLAlchemy models, and JWT auth with bcrypt hashing.
- Wrote 2 pytest integration suites with TestClient; deployed to Railway and Vercel with CORS and env config.

StatDiamond | Baseball Analytics Platform Deployed on Vercel
Python, FastAPI, PostgreSQL, SQLAlchemy, React, TypeScript, Supabase, JWT, pybaseball
- Built a baseball analytics app serving live standings and stats for all 30 MLB teams through a FastAPI REST API.
- Designed a normalized PostgreSQL schema of 3 relational models with full CRUD endpoints, JWT auth, and role-based access.

EDUCATION

University of Colorado Boulder | B.S. Computer Science (Post-Baccalaureate), GPA: 4.0 In Progress

University of Colorado Boulder | B.S. Business Administration, Leeds School of Business 2020
Chancellor's Achievement Scholar | Study Abroad, University of Westminster, London

Actualize Coding Bootcamp | Full-Stack Web Development (Ruby on Rails, JavaScript, React) 2024

CERTIFICATIONS

FINRA Series 3, 7, 63, 65 | Certified Blockchain & Digital Assets (CBDA) | Google Project Management | Tableau